

The PneuCarriage Project: A Multi-Centre Comparative Study to Identify the Best Serotyping Methods for Examining Pneumococcal Carriage in Vaccine Evaluation Studies

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Evidence abstract

BACKGROUND The pneumococcus is a diverse pathogen whose primary niche is the nasopharynx. Over 90 different serotypes exist, and nasopharyngeal carriage of multiple serotypes is common.

Understanding pneumococcal carriage is essential for evaluating the impact of pneumococcal vaccines.

Traditional serotyping methods are cumbersome and insufficient for detecting multiple serotype carriage, and there are few data comparing the new methods that have been developed over the past decade.

We established the PneuCarriage project, a large, international multi-centre study dedicated to the identification of the best pneumococcal serotyping methods for carriage studies.

METHODS AND FINDINGS Reference sample sets were distributed to 15 research groups for blinded testing.

Twenty pneumococcal serotyping methods were used to test 81 laboratory-prepared (spiked) samples.

The five top-performing methods were used to test 260 nasopharyngeal (field) samples collected from children in six high-burden countries.

Sensitivity and positive predictive value (PPV) were determined for the test methods and the reference method (traditional serotyping of >100 colonies from each sample).

For the alternate serotyping methods, the overall sensitivity ranged from 1% to 99% (reference method 98%), and PPV from 8% to 100% (reference method 100%), when testing the spiked samples.

Fifteen methods had 70% sensitivity to detect the dominant (major) serotype, whilst only eight methods had 70% sensitivity to detect minor serotypes.

For the field samples, the overall sensitivity ranged from 74.2% to 95.8% (reference method 93.8%), and PPV from 82.2% to 96.4% (reference method 99.6%).

The microarray had the highest sensitivity (95.8%) and high PPV (93.7%).

The major limitation of this study is that not all of the available alternative serotyping methods were included.

CONCLUSIONS Most methods were able to detect the dominant serotype in a sample, but many performed poorly in detecting the minor serotype populations.

Microarray with a culture amplification step was the top-performing method.

Results from this comprehensive evaluation will inform future vaccine evaluation and impact studies, particularly in low-income settings, where pneumococcal disease burden remains high.

Document metadata

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Evidence checklist

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2. Over 90 different serotypes exist, and nasopharyngeal carriage of multiple serotypes is common.
3. Understanding pneumococcal carriage is essential for evaluating the impact of pneumococcal vaccines.
4. Traditional serotyping methods are cumbersome and insufficient for detecting multiple serotype carriage, and there are few data comparing the new methods that have been developed over the past decade.
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6. METHODS AND FINDINGS Reference sample sets were distributed to 15 research groups for blinded testing.
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9. Sensitivity and positive predictive value (PPV) were determined for the test methods and the reference method (traditional serotyping of >100 colonies from each sample).
10. For the alternate serotyping methods, the overall sensitivity ranged from 1% to 99% (reference method 98%), and PPV from 8% to 100% (reference method 100%), when testing the spiked samples.
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13. The microarray had the highest sensitivity (95.8%) and high PPV (93.7%).
14. The major limitation of this study is that not all of the available alternative serotyping methods were included.
15. CONCLUSIONS Most methods were able to detect the dominant serotype in a sample, but many performed poorly in detecting the minor serotype populations.
16. Microarray with a culture amplification step was the top-performing method.
17. Results from this comprehensive evaluation will inform future vaccine evaluation and impact studies, particularly in low-income settings, where pneumococcal disease burden remains high.

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